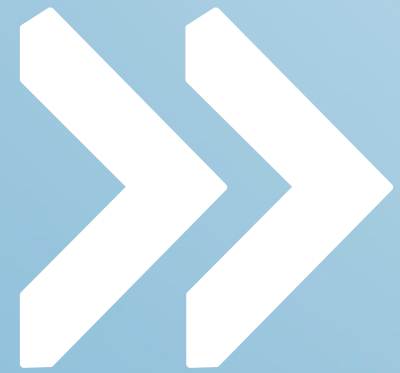
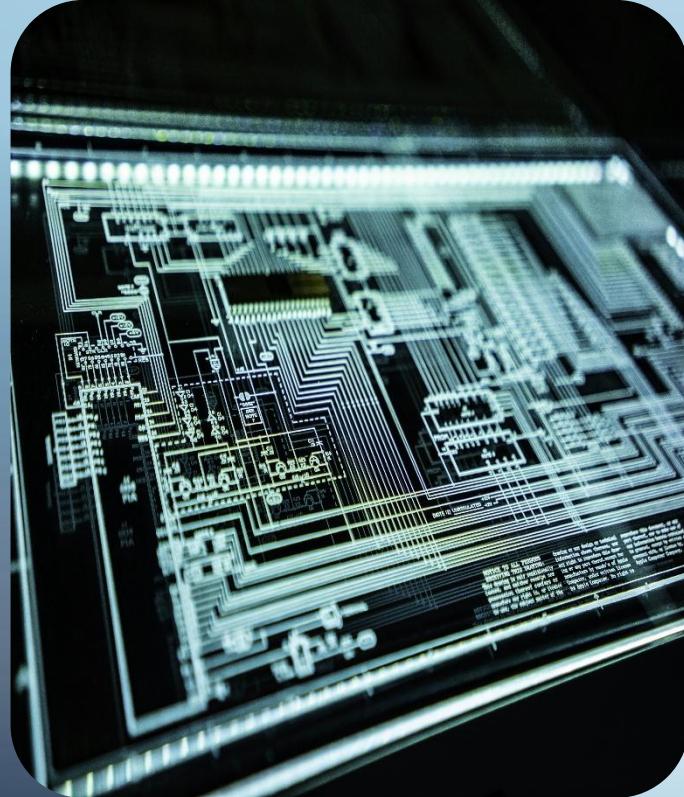


ENVIRONMENTAL BLACK BOX RECORDER



**IoT Based Micro-Project by
Team SOILSYNC SYSTEMS – ACE Engineering College**



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SoilSync Systems – ACE Engineering College

PROBLEM STATEMENT

Lack of Real-Time Multi-Parameter Environmental Monitoring

In many environments—industries, farms, homes, and storage areas—critical conditions are identified only after damage occurs because current monitoring solutions are single-parameter and lack real-time alerts.

They also do not save past data, making it difficult to understand what caused an incident.

This creates the need for a unified IoT-based system that continuously monitors multiple environmental factors, detects abnormalities instantly, and records data for future investigation.

WHAT IS THE ENVIRONMENTAL BLACK BOX RECORDER?

The System That Monitors & Protects Your Environment

The Environmental Black Box Recorder is a multi-sensor monitoring system designed to continuously track critical environmental parameters such as temperature, humidity, gas levels, moisture, water presence, light intensity, and tilt.

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It acts as an intelligent safety unit that identifies hazardous conditions early and records data for analysis.

The ESP32 microcontroller enables real-time processing and wireless monitoring, making the system powerful yet compact.





Continuous Monitoring for Environmental Safety

Modern environments — homes, industries, storage spaces, and agriculture setups — require reliable monitoring to detect harmful changes.

This project provides a unified solution by integrating multiple sensors into a single device, capable of recording, analyzing, and alerting abnormal environmental conditions. Inspired by the idea of an aircraft “black box,” it stores crucial environmental information and aids in preventive safety.

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THE
PURPOSE OF
THE
PROJECT



THE CORE PRINCIPLE OF OUR SYSTEM



Multi-Sensor Intelligence Powered by ESP32

The ESP32 collects readings from different sensors and evaluates them using predefined safety thresholds.

If unusual activity is detected — like gas leaks, temperature rise, water level changes, or sudden tilt — the system instantly triggers alarms using LEDs and a buzzer.

This design ensures rapid response to environmental threats while logging real-time data for users to monitor.



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SENSORS USED IN OUR SYSTEM



Temperature & Humidity Monitoring (DHT11)

Measures environmental temperature and humidity to detect heat rise or moisture imbalance.

Light Intensity Detection (LDR Sensor)

Monitors brightness changes, useful for night mode activation and intrusion indication.

Gas & Smoke Measurement (MQ-2 Sensor)

Detects flammable gases, smoke, and air-quality hazards in real time.



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Soil Moisture Sensor

Monitors environmental or soil moisture by measuring conductivity between probes.

Used to detect dry/wet conditions, useful in agriculture or leak detection surfaces.

Water Level Sensor

Detects the presence or change in water levels using conductive traces.

Triggers warnings during water leakage, flooding, or overflow conditions.

Tilt Switch Sensor

A simple electromechanical sensor that detects tilt, vibration, or sudden movement.

Helps identify disturbances, shock events, or device orientation changes similar to a black-box recorder.



OUR SYSTEM

COMPONENTS

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ESP32 Microcontroller

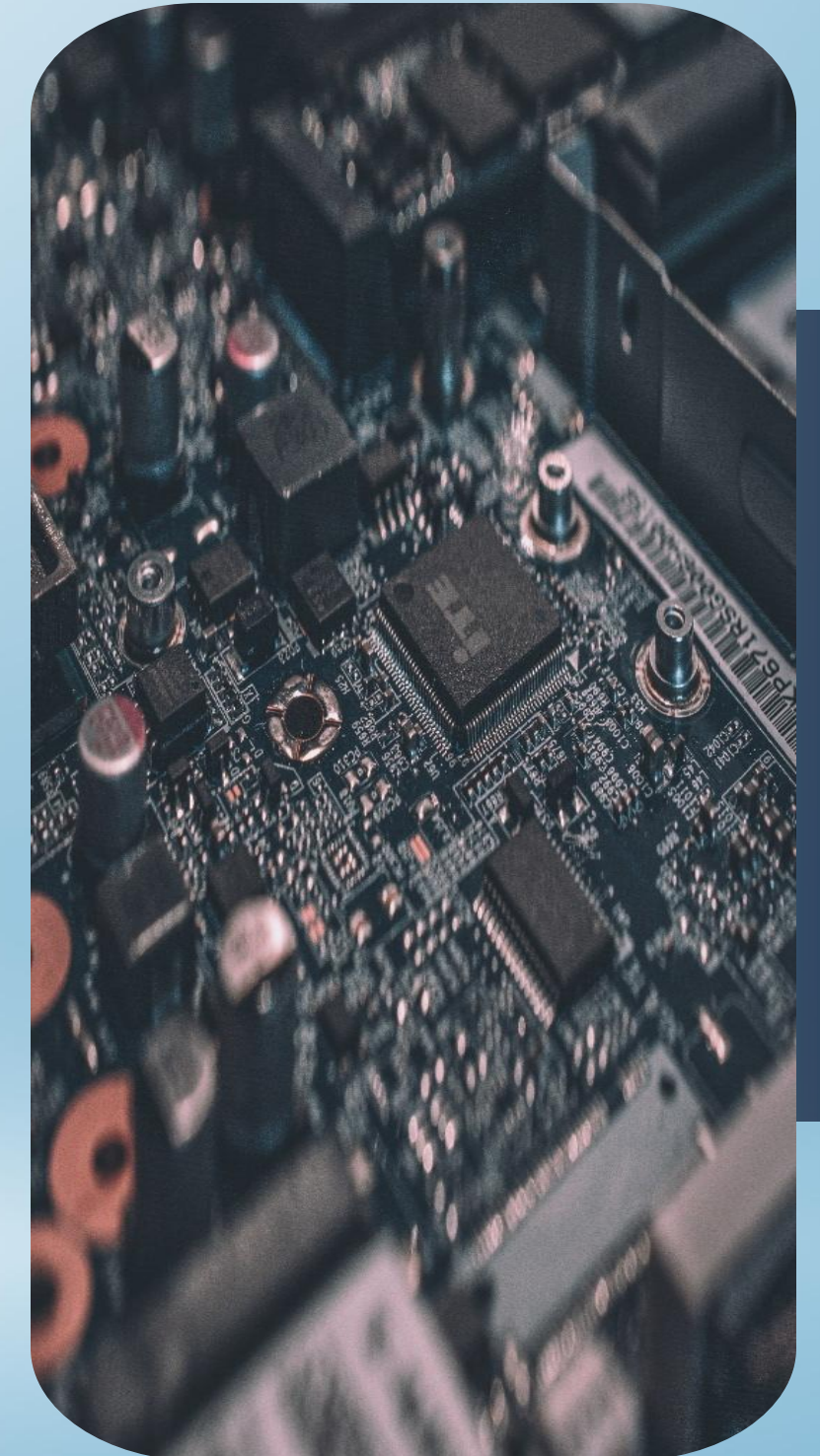
Central processing unit
managing sensors, alerts,
and Wi-Fi dashboard.

Buzzer & LEDs

Provides immediate alert
responses for critical
sensor readings.

Sensors + Breadboard Setup

Unified integration of all
sensors enabling accurate
monitoring.



How The System Works

Understanding the Functioning of the Black Box Recorder

Real-Time Sensor Data Collection

All sensors record environmental values continuously.
These readings form the core input to the system.

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Alert Triggering Mechanisms

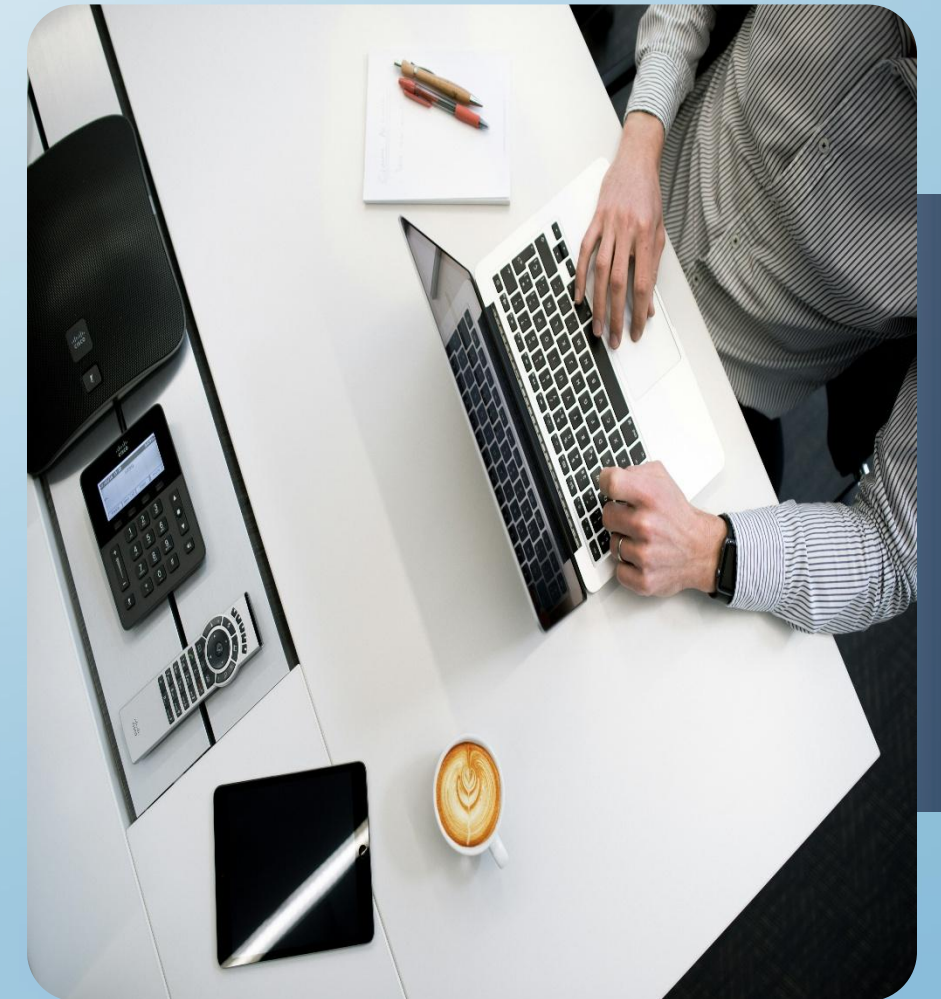
Buzzer and LEDs respond to critical events.
Visual and audio cues ensure quick user awareness.

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Live Monitoring via Wi-Fi Dashboard

ESP32 hosts a real-time web interface.
Users can view current values from any connected device.

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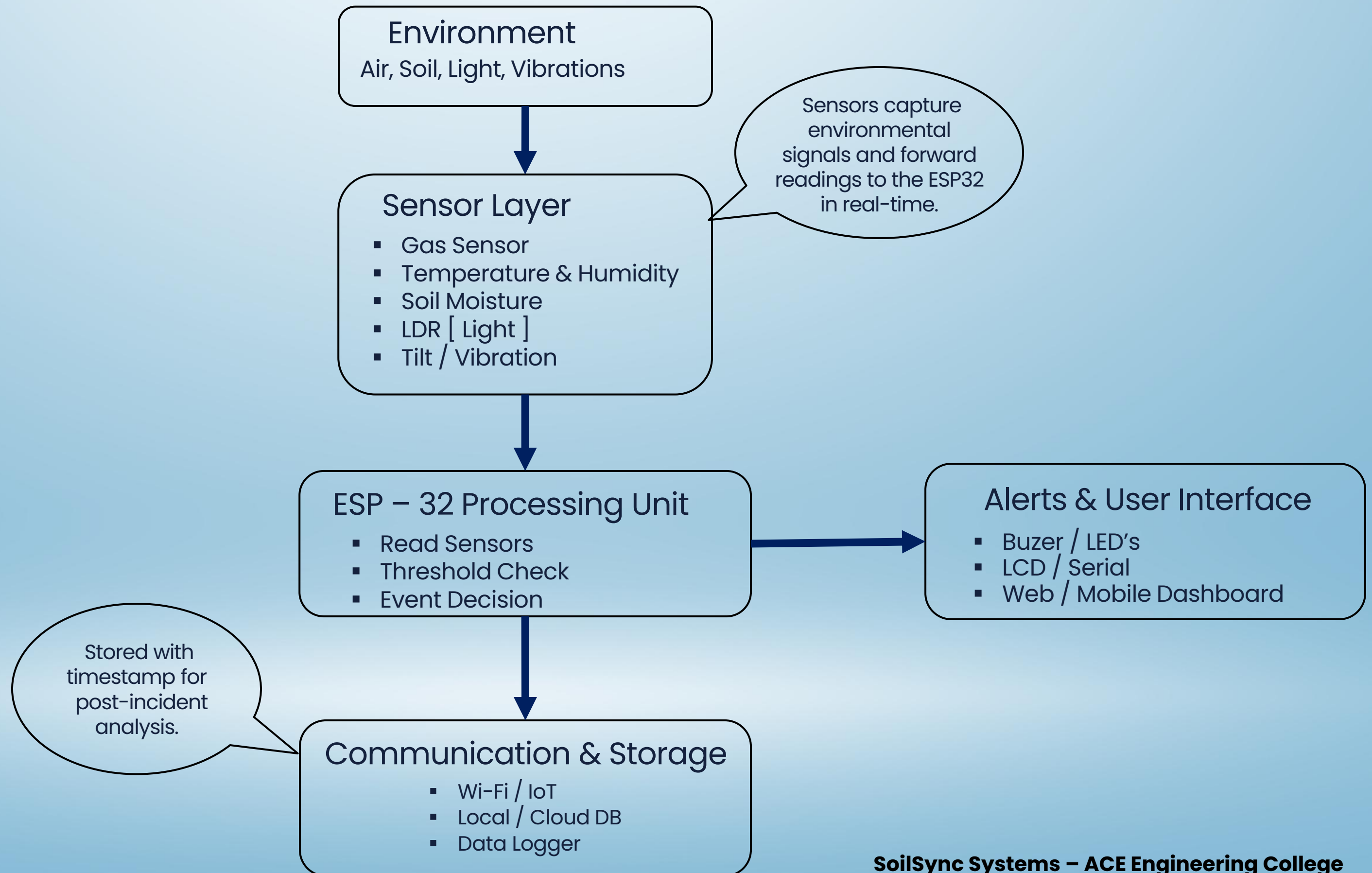


Processing & Threshold Evaluation

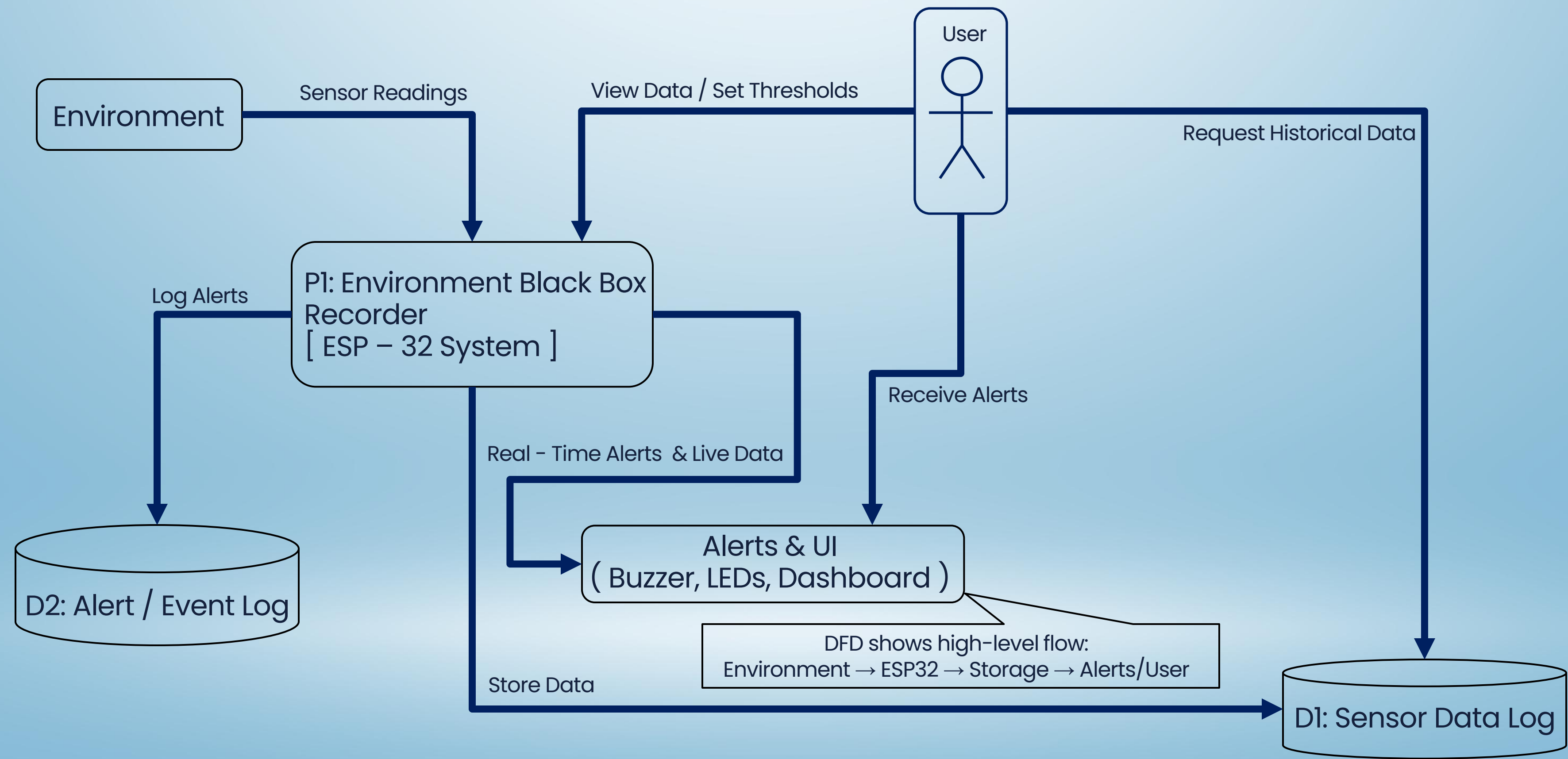
ESP32 compares sensor inputs against predefined safety limits.
If danger is detected, alert mechanisms activate instantly.

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SYSTEM ARCHITECTURE



DATA FLOW DESIGN



SYSTEM FEATURES

Intelligent Environmental Surveillance

Continuous Multi-Parameter Tracking

Monitors temperature, humidity, gas levels, light, moisture, and tilt 24/7. Ensures maximum visibility into environmental changes.

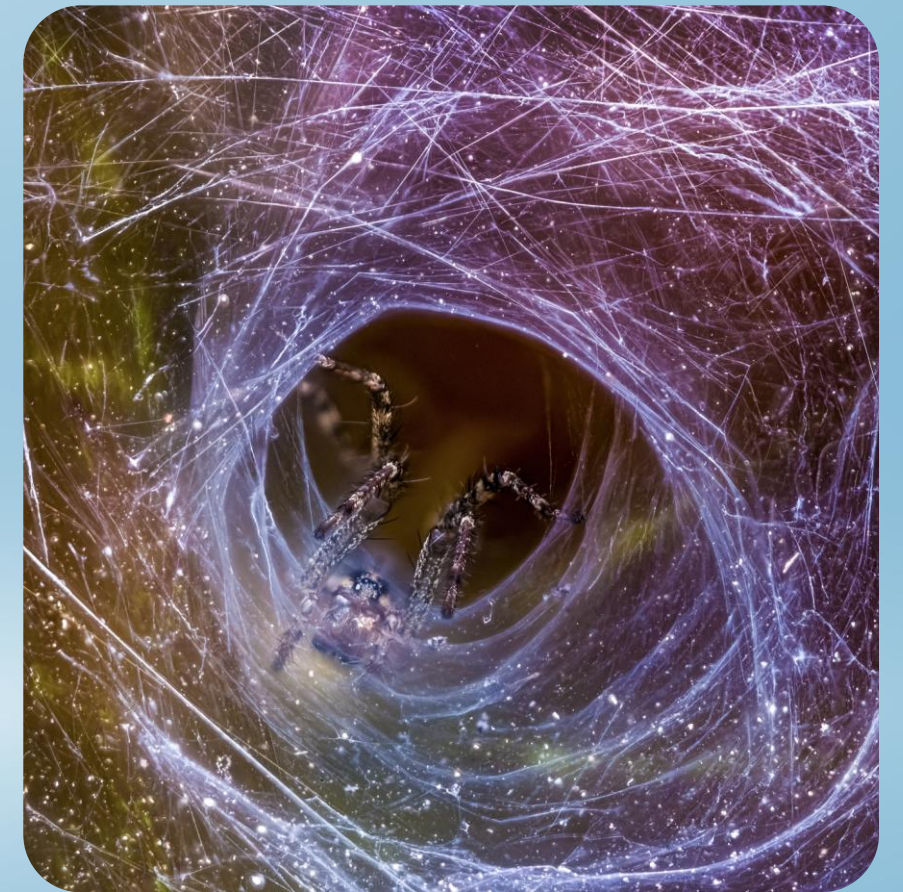
Instant Hazard Alerts

The system reacts quickly to abnormal sensor values. Critical warnings prevent accidents and enhance safety.

Compact IoT Architecture

ESP32's built-in Wi-Fi capabilities enable seamless connectivity. Suitable for homes, greenhouses, labs, and industrial zones.

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APPLICATIONS OF OUR SYSTEM

Where the Environmental Black Box Makes an Impact



Smart Homes

Detects gas leaks, water leakage, temperature rise, and unsafe environmental shifts.

Improves household safety and automation.



Agriculture & Farming

Tracks soil moisture, humidity, and temperature trends.

Helps optimize irrigation and protect crop environments.



Industrial & Laboratory Safety

Monitors hazardous gases, humidity variations, and heat exposure.

Supports safe operation in sensitive workspaces.



REAL-WORLD IMPACT

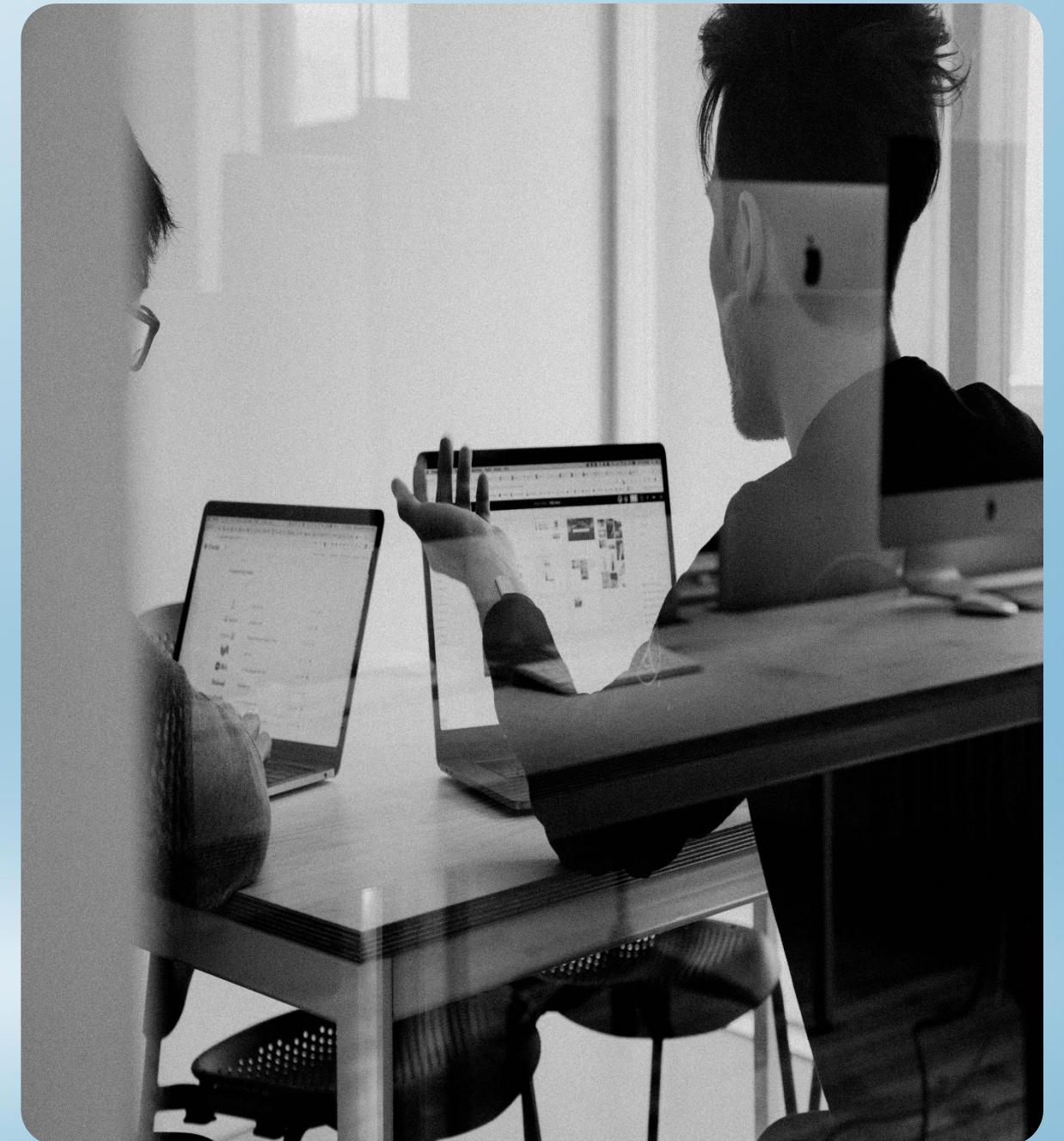
Enhancing Safety Through Smart Environmental Insights

The Environmental Black Box Recorder strengthens safety by identifying harmful conditions before they escalate.

It supports IoT-based monitoring, enabling users to view live data and respond quickly during emergencies.

Its ability to monitor multiple factors makes it valuable for smart cities, industries, research labs, and sustainable environments.

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THANK YOU

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